Tran Thu An (An) Phung

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Technical Skills

Languages: C++, Java, Python, JavaScript/TypeScript, React (Web + Native), FastAPI, Spring Boot.
Testing & QA: JUnit, Postman,CI/CD Test Integration.
Tools: Docker, GitHub Actions, Nginx, Linux, AWS (EC2, S3), SQL Server, PostgreSQL.
Practices: QA Methodologies, Test Planning, Defect Documentation, Agile, REST API, Cross-browser Testing.

Technical Experiences

Backend Engineering Intern May 2024 - Aug 2024
Thang Long Software Solution *Hanoi, Vietnam*

- Optimized queries and redesigned **PostgreSQL** schemas for 700+ administrative units, cutting query response time by 30% and enabling future ORM integration.
- Automated **data extraction** and **reporting** in **C++,** reducing manual workload across multiple departments.
- Supported defect investigation and debugging of backend components, documenting fixes and contributing to maintainable code.
- Gained exposure to socket-based communication (**kbmMW over TCP/IP**), broadening backend integration skills beyond REST APIs.
- Participated in **Agile sprints and architectural discussions**, gaining exposure to large-scale backend design.

Technical Projects

QuickNote - AI Document Q&A Platform Jul 2025 - Present

- Built a full-stack AI system for document upload and contextual Q&A using an **OpenAI RAG pipeline.**
- Developed responsive UI with **React and TypeScript**; implemented API-driven file management and chat features.
- Implemented secure backend services with **FastAPI, JWT auth, PostgreSQL, Pinecone, AWS S3**, handling data ingestion, embeddings, and contextual Q&A; tested backend APIs with **Postman** to validate authentication, file handling, and chat workflows.
- Deployed in a modular, production-ready architecture using **Docker and Nginx**, ensuring scalability, maintainability, and reliable cloud integration.

Virtual Pet Game Nov 2024

- Built and tested a **Java** desktop game simulating pet-care routines, with unit tests written in **JUnit**.
- Applied **object-oriented programming (OOP)** principles and software design patterns to ensure a scalable and maintainable codebase.
- Developed the interface using **Java Swing**, delivering a smooth and interactive player experience.

Other Experiences

Undergraduate Teaching Assistant Sep 2025 - Present
University of British Columbia *Vancouver, BC*

- Assisted in teaching MATH 100 - Differential Calculus with Applications to over 100 students.
- Coordinated with instructors to align assessment, feedback, and student support; strengthened technical communication.
- Conducted weekly tutorial sessions, clarifying mathematical concepts, and provided assignment feedback.

Education

University of British Columbia *Expected Graduation: May 2028*
Bachelor of Science (B.Sc.) in Computer Science *GPA: 90.4*

Coursework: Applied Machine Learning, Relational Database, Software Engineering, Operating Systems, Object-Oriented Design, Algorithms, Software Construction.
Awards: Science's Dean Honour List